

LPS-1: Polygon Grant Application Narrative

Applicant: XXXIII Working Group (FTHTrading) **Protocol:** Literary Publishing Standard (LPS-1) **Network:** Polygon Mainnet (Chain ID 137) **Date:** February 2026 **Contact:** xxxiii.io | github.com/FTHTrading/LPS-1-Reference-Implementation

1. Executive Summary

LPS-1 is an open, deterministic provenance standard for verifiable digital manuscripts, deployed live on Polygon Mainnet.

The protocol establishes cryptographic proof-of-origin, content integrity, and immutable authorship binding using SHA-256 hashing, Merkle tree commitments, IPFS pinning, and on-chain anchoring — without relying on any trusted intermediary. It is fully specified (RFC-style, 14 sections), fully deployed (7 verified contracts, zero upgradeability), fully tested (293 tests across 7 suites), and fully open (MIT license, CC BY 4.0 specification).

LPS-1 is public goods infrastructure. It is not a platform, marketplace, or publishing tool. It is a standard for proving that a specific person wrote a specific work at a specific time — using mathematics, not intermediaries.

2. Problem Statement

Digital works have no standardised mechanism for cryptographic proof-of-origin.

Authors cannot independently prove when a work was written, that its contents have not been altered, or that a specific individual authored it — without relying on centralised intermediaries. Existing publishing infrastructure depends on institutional trust: publishers, distributors, and platform operators serve as de facto authorities on authorship and content integrity.

This problem is compounded by three converging trends:

AI-generated content proliferation. As generative AI produces text indistinguishable from human-authored work, the need for deterministic, verifiable authorship proof becomes critical. No existing standard addresses this.

Platform dependency risk. Authors who rely on Amazon, Google, or any single platform for provenance are subject to arbitrary policy changes, deplatforming, and corporate dissolution. Authorship proof must be platform-independent.

Institutional trust erosion. Copyright registries, ISBN databases, and publisher attestations are centralised points of failure. They cannot be independently verified without institutional cooperation.

The problem is structural. It is not solved by digitisation, e-book platforms, or blockchain-based marketplaces. These tools replicate the same trust model in digital form. What is required is an open, deterministic standard that anyone can implement and anyone can verify.

3. Solution: The LPS-1 Protocol

LPS-1 defines a deterministic protocol for literary provenance with six independently verifiable layers:

Layer	Function	Verification Method
Filesystem	Canonical file ordering and version control	<code>git log</code> , directory inspection
Git	Commit-level authorship history with GPG signatures	<code>git verify-commit</code>
SHA-256	FIPS 180-4 cryptographic fingerprint at word/paragraph/chapter level	<code>sha256sum</code>
Merkle Trees	Per-content-type provenance commitments (manuscript, audio, image, artifact)	Root comparison against on-chain anchor
IPFS	Content-addressed decentralised storage	CID resolution
Polygon	Immutable on-chain anchor with block timestamp	Direct RPC read, Polygonscan verification

Additionally, cross-chain timestamping via Bitcoin (OpenTimestamps) provides independent temporal attestation on the most secure public blockchain.

Key protocol properties:

- **Deterministic builds.** Given identical source files, the protocol produces identical hashes. No randomness, no environment dependency.
- **Forward-only lifecycle.** Editions move from Draft → Anchored → Frozen. Frozen editions cannot be modified or deleted. State transitions are enforced on-chain.
- **Zero upgradeability.** All seven smart contracts are non-upgradeable by design. No proxy pattern, no admin key, no governance override.
- **Client-side verification.** Any user can verify any claim using only a blockchain RPC endpoint and publicly available source files. No backend, no API key, no trusted service.

4. What Is Already Live

LPS-1 is not a proposal. It is a deployed, production system on Polygon Mainnet.

Metric	Value
Smart contracts deployed and verified	7 / 7
Test coverage	293 tests across 7 suites
Reference implementation verification tests	58 (37 contract + 21 pipeline)
Compliance level achieved	Level 5 — Fully Observable
Reference implementations frozen on-chain	2 (The 2,500 Donkeys, Private Placement Programs)
Merkle trees constructed and anchored	6 (manuscript, audio, image, artifact, prompt, combined)
Cross-chain timestamps	Polygon + Bitcoin (OpenTimestamps)
Architecture Decision Records	6 documented
Specification status	Informational, RFC-style, 14 sections
Research paper	Published, DOI-indexed, open access
Client-side verification dashboard	Live at xxxiii.io
Upgrade mechanism	None (non-upgradeable by design)

License	MIT (infrastructure), CC BY 4.0 (specification)
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Deployed contracts (all verified on Polygonscan):

Contract	Address	Function
LiteraryAnchor	0x97f456300817eaE3B40E235857b856dFE8bba90	Edition anchoring, hash/CID storage
KernelV2	0xca9F6604A9b498DB31d113836E2957c0a9aAE037	Edition lifecycle, freeze enforcement
AuthorIdentity	0xB9ffa688A8Bb332221030BbBE46bE5bF03323170	ECDSA-bound identity, pseudonym registry
RoyaltyRouter	0x44169829489d70aaecbf845870652871C65fC461	ERC-2981 routing
EditionNFT	0x9e9Cc1486bf440Bd9eAaD947958524Aaed3f8b0	ERC-721 edition tokens
StoryNFT	0xD67e537DbA1236f802432cbDD30Fec3f6D38e7E3	ERC-721 story tokens
Kernel (v1)	0x511c653fC0F450ba41C42A89A3125CcBf2eFE8ae	Legacy kernel (superseded)

5. Why Polygon

Polygon Mainnet was selected as the primary settlement layer based on a structured cost-benefit analysis (documented in ADR-0006):

Factor	Assessment
Transaction cost	< \$0.01 per anchor operation
Finality	~2 second block times
EVM compatibility	Full Solidity 0.8.19 support, standard tooling
Verification	Polygonscan source verification, public RPC endpoints
Ecosystem	Established validator set, institutional adoption
Longevity	Polygon Labs backing, Ethereum-aligned roadmap

LPS-1 is native Polygon infrastructure. Every contract, every anchor, every verification read operates on Polygon Mainnet. The protocol's live telemetry dashboard at xxxiii.io reads directly from Polygon RPC — no intermediary backend, no caching layer, no API abstraction.

6. Grant Request Scope

This grant funds the transition from single-implementation protocol to multi-implementation standard. Four phases:

Phase II — Multi-Author Support

- Co-author identity binding with shared ECDSA governance
- Delegation support for institutional publishers
- Shared edition lifecycle management
- **Deliverable:** Updated AuthorIdentity contract + specification amendment

Phase IV — zk-Proof Inclusion

- Zero-knowledge Merkle inclusion proofs for privacy-preserving verification
- Proof generation library (TypeScript + Solidity verifier)
- **Deliverable:** zk-SNARK verifier contract on Polygon, npm package

Phase V — Institutional API + SDK

- REST verification endpoints (OpenAPI 3.x)
- TypeScript SDK for programmatic anchoring and verification
- Integration guides for libraries, archives, and publishers
- **Deliverable:** Published npm package, hosted API documentation

Independent Security Review

- Third-party audit of all seven deployed contracts
 - Formal verification of critical state transitions (freeze, anchor)
 - **Deliverable:** Published audit report, remediation documentation
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7. Budget Allocation

Category	Allocation	Scope
Protocol Engineering	40%	Multi-author contracts, zk-proof system, SDK development
Security Audit	25%	Third-party contract audit, formal verification
Documentation & SDK	20%	OpenAPI specification, integration guides, npm packaging
Ecosystem Outreach	15%	Implementor onboarding, academic partnerships, conference presentation

Budget is structured for accountability. Each phase has discrete deliverables with on-chain verification. Contract deployments are publicly verifiable. All code is MIT-licensed and pushed to the public repository.

8. Public Goods Impact

LPS-1 is public goods infrastructure. It is designed to be used by anyone, verified by anyone, and implemented by anyone — without permission, payment, or platform dependency.

Open Reproducibility

Every claim made by the protocol is independently reproducible. Given the source files and a Polygon RPC endpoint, any party can reconstruct every hash, verify every Merkle root, and confirm every on-chain anchor. No trusted service required.

Platform-Independent Verification

Authors using LPS-1 do not depend on any platform, publisher, or service provider to prove authorship. The proof exists on-chain, is readable by anyone, and persists independently of the author's continued participation.

AI Transparency Layer

In an era of generative AI, LPS-1 provides a mechanism for authors to cryptographically distinguish human-authored works from AI-generated content — not through claims, but through deterministic, timestamped provenance chains that predate the work's publication.

Cultural Infrastructure

Literary works are cultural artifacts. Their provenance should be as verifiable as financial transactions. LPS-1 treats literary publishing with the same cryptographic rigour that blockchain applies to value transfer.

Standards-Grade Design

LPS-1 is not a product. It is a standard. It has:

- A formal specification (RFC-style, 14 sections)
 - Compliance levels (LO–L5 with published evidence requirements)
 - A governance model (author-led → working group transition)
 - A published research paper (DOI-indexed, open access)
 - A conformance registry for external implementations
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9. Adoption Strategy

Current State

- 1 production implementation (XXXIII, Level 5)
- 2 frozen reference implementations on-chain
- Public "Become an Implementor" onboarding path at xxxiii.io
- Open conformance registry accepting external submissions

Growth Path

1. **Implementor onboarding.** Five-step process published at xxxiii.io: choose compliance level, clone repo, run verifier, deploy anchor, submit conformance statement.
2. **Academic adoption.** Research paper published with DOI. Protocol suitable for digital humanities grants (NEH, Mellon Foundation, IMLS).

3. **Institutional integration.** Phase V delivers REST API for library and archive integration. Enables institutional verification without requiring blockchain expertise.
4. **Cross-ecosystem reach.** Phase III (Ethereum L1 mirror) extends protocol to settlement-layer finality, broadening the implementor base beyond Polygon.

Success Metrics

Metric	6 Months	12 Months
External implementations	1+	3+
Specification amendments	1 (multi-author)	2 (multi-author, zk)
Npm package downloads	—	500+
Security audit	Initiated	Published
Governance transition	—	Working group formed

10. Team

Role	Name	Responsibility
Protocol Author	Kidd James	Specification, contract architecture, reference implementation
Specification Steward	XXXIII Working Group	Specification maintenance, amendment review
Reference Implementation	FTHTrading	Repository management, CI/CD, test infrastructure
Security Review	Independent (2026)	Contract audit, formal verification

11. References

Resource	Link
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Protocol Site	xxxiii.io
LPS-1 Specification	spec/LPS-1.md
LPS Protocol Stack	spec/LPS-STACK.md
Compliance Matrix	spec/COMPLIANCE.md
Governance Model	spec/GOVERNANCE.md
Roadmap	docs/ROADMAP.md
Funding Brief	docs/FUNDING_BRIEF.md
Research Paper	DOI 10.5281/zenodo.18646886
Public Repository	GitHub
LiteraryAnchor Contract	Polygonscan
KernelV2 Contract	Polygonscan